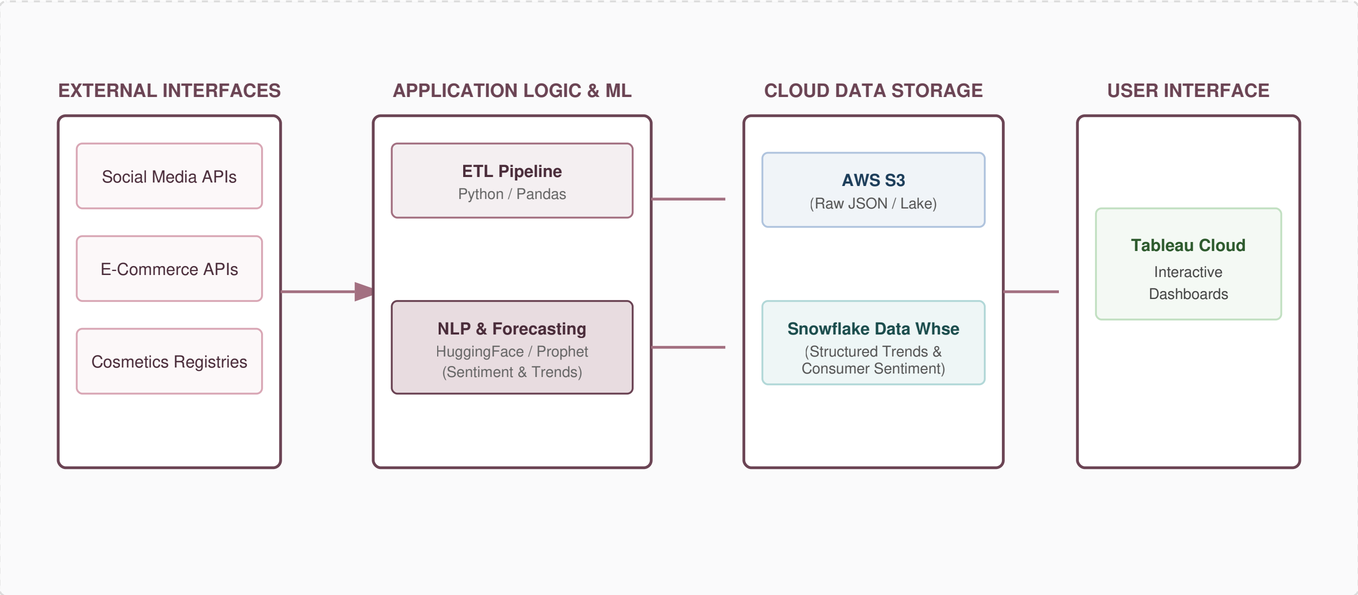
Date	7th July 2026	Maximum Marks	4 Marks
Team ID	XXXXXX	Project Name	**Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau**

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Technical Architecture Diagram



Architectural Demarcation & Design Considerations:

- **Processes & Logic:** Scheduled ingestion pipelines pull unstructured sentiment data and structured brand performance metrics.
- **Infrastructural Demarcation:** Data collection, analytics transformations, and ML staging run on secure Cloud infrastructure (AWS/Snowflake).
- **External Interfaces:** Third-party REST APIs facilitate data extraction from verified e-commerce review structures and social channels.
- **Data Storage:** Raw logs use object storage, while curated, structured trend schemas reside in a cloud data warehouse optimized for Tableau connectivity.

Table-1: Technology Stack Details

S.No	Component	Description	Technology
1	**User Interface**	Interactive executive dashboards and trend explorers providing filters for consumer segments, categories, and ingredient sentiments.	Tableau Cloud / Embedded Tableau Web UI
2	**Application Logic-1**	Automated ETL pipelines responsible for continuous data extraction, schema validation, normalization, and cleansing of cosmetic review datasets.	Python (Pandas, NumPy, Requests)
3	**Application Logic-2**	Data orchestration layer managing workflows, dependency scheduling, and execution tasks for the data pipeline.	Apache Airflow / Prefect
4	**Application Logic-3**	Data transformation and modeling layer to curate optimized, aggregate-level analytical views suited for rapid BI dashboard queries.	dbt (Data Build Tool)
5	**Database**	Local metadata repository and execution logging store utilized during pipeline development and local integration phases.	PostgreSQL (v15+)
6	**Cloud Database**	High-performance cloud data warehouse housing refined trend matrix tables, ingredient profiles, and historic consumer sentiment vectors.	Snowflake Data Warehouse
7	**File Storage**	Object storage repository serving as a data lake for raw JSON API payload snapshots and unstructured web-scraped historical reviews.	AWS S3 (Simple Storage Service)
8	**External API-1**	Social graph and marketing analytics endpoints utilized to retrieve trending hashtags, mention volumes, and viral beauty product engagement indices.	TikTok Research API / Instagram Graph API
9	**External API-2**	E-commerce API integrations used to fetch public review data, rating breakdowns, and verified consumer product descriptions.	RapidAPI / Brand-Specific Merchant Endpoints
10	**Machine Learning Model**	Natural Language Processing (NLP) pipeline evaluating review sentiments, plus time-series forecasting predicting next-quarter cosmetic trend cycles.	HuggingFace Transformers (RoBERTa) & Meta Prophet

S.No	Component	Description	Technology
11	**Infrastructure (Server / Cloud)**	Fully containerized pipeline infrastructure deployed via cloud container orchestrators to guarantee deterministic environments. **Local Server Configuration:** Docker Desktop, Minikube, 16GB RAM, Core i7. **Cloud Server Configuration:** AWS ECS / EKS (Elastic Kubernetes Service), Tableau Managed SaaS Environment.	AWS Cloud, Docker, Kubernetes

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	Core open-source libraries enabling data ingestion, mathematical array manipulation, machine learning execution, and orchestration.	Python, Pandas, Scikit-learn, PyTorch, Apache Airflow
2	**Security Implementations**	End-to-end data security ensuring data is encrypted at rest and in transit. Granular data access enforced using role-based dashboard configurations.	AES-256 Encryption, TLS 1.3, AWS IAM, Tableau Row-Level Security (RLS)
3	**Scalable Architecture**	Decoupled multi-tier cloud data architecture where computing layers scale dynamically independently from storage tiers to handle multi-gigabyte text datasets.	Snowflake Virtual Warehouses, AWS Auto-scaling Groups
4	**Availability**	High availability backed by automated cloud database replicas spanning multiple availability zones and managed BI cloud availability guarantees.	AWS Multi-AZ Deployment, Tableau Cloud High Availability Architecture
5	**Performance**	Aggressive query acceleration using optimized workbook data hyper-extracts, data warehouse result-set caching, and pre-computed analytical views.	Tableau Hyper Extracts, Snowflake Query Cache & Materialized Views

References

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